Business Question and Visualization Report

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

Sample

Project: Global Trends of AI Adoption in Education

1. How is AI adoption distributed across high, low, and moderate adoption levels?

Visualization: The bar chart compares the number of records in each adoption category. Low adoption is the largest category (680), followed by moderate adoption (473) and high adoption (207).

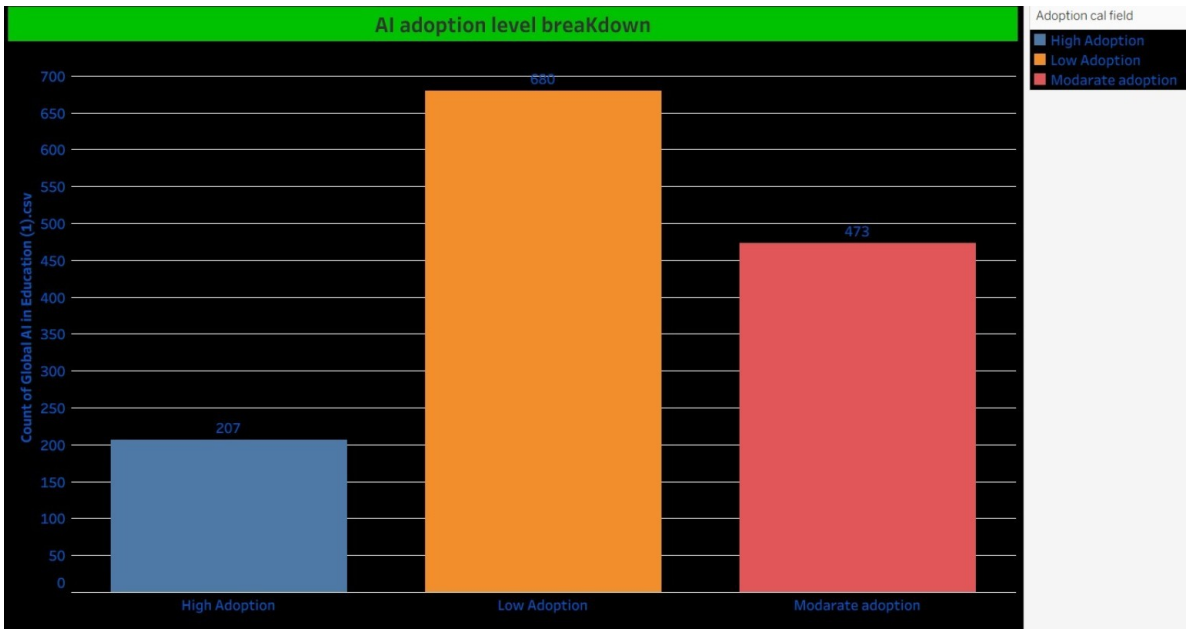


Figure: AI adoption level breakdown

2. How does school AI adoption compare across world regions?

Visualization: The regional pie chart shows a broadly similar distribution across the displayed regions, with values clustered around 29.25–29.53.

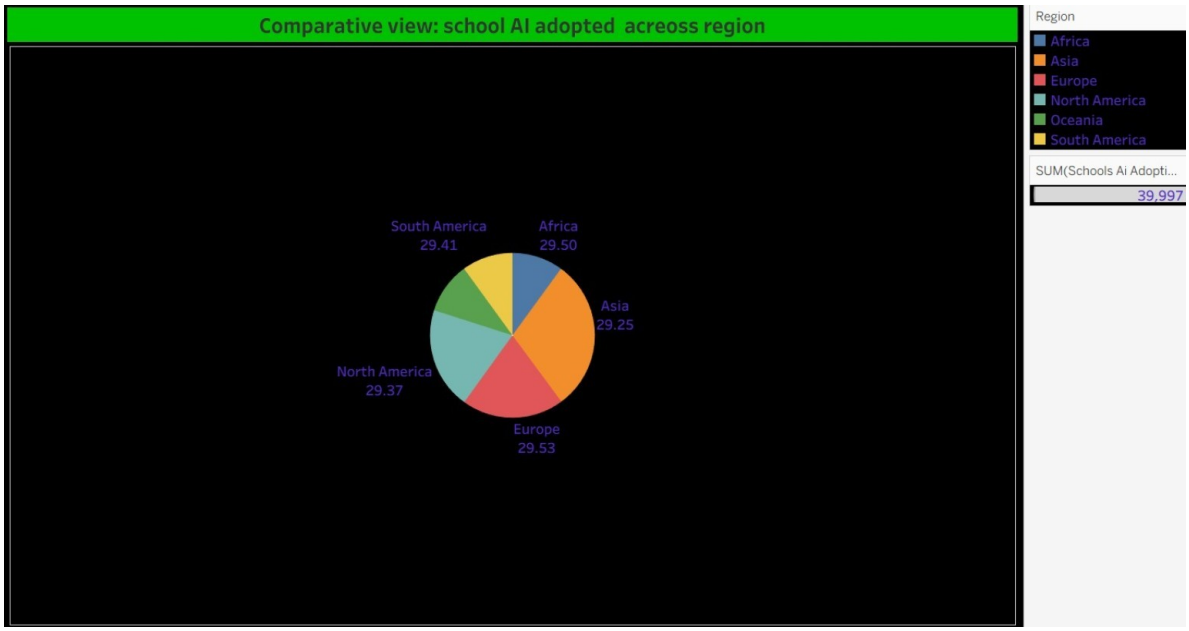


Figure: Comparative view of school AI adoption across regions

3. How has AI usage changed in rural and urban communities over time?

Visualization: The mirrored bar chart shows an overall upward trend in both settings from 2015 to 2026. Urban usage is higher than rural usage throughout the displayed period.

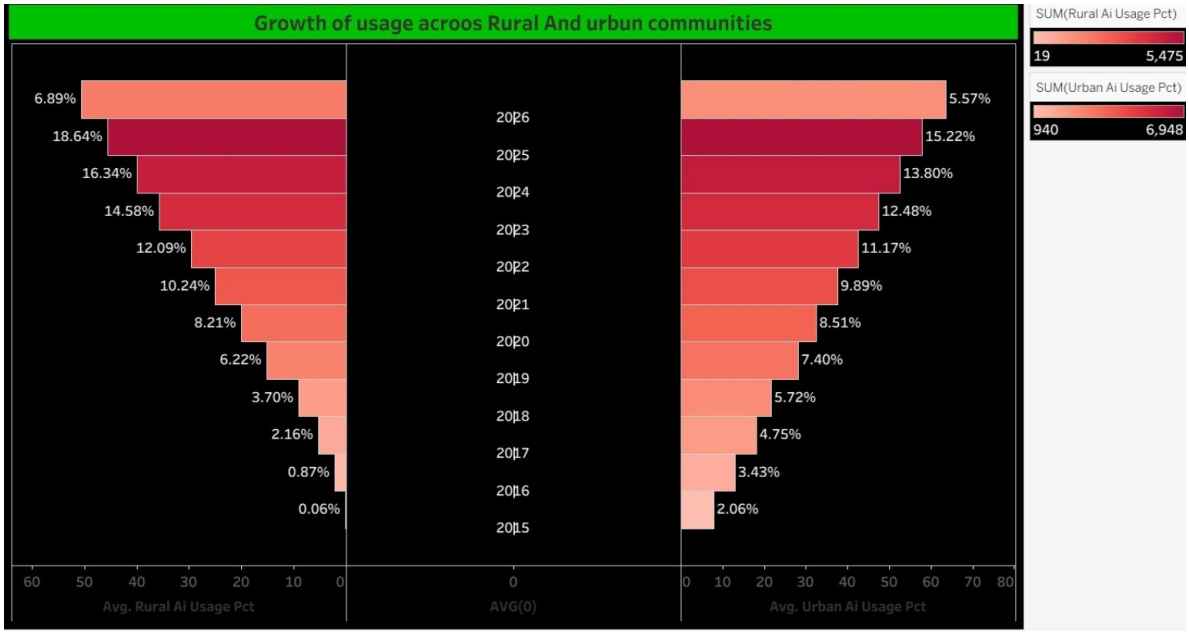


Figure: Growth of AI usage across rural and urban communities

4. How has AI adoption in education developed from 2015 to 2026?

Visualization: The three trend lines show steady growth in student, school, and teacher AI usage over the period. By 2026, the displayed values reach approximately 60% for each measure.

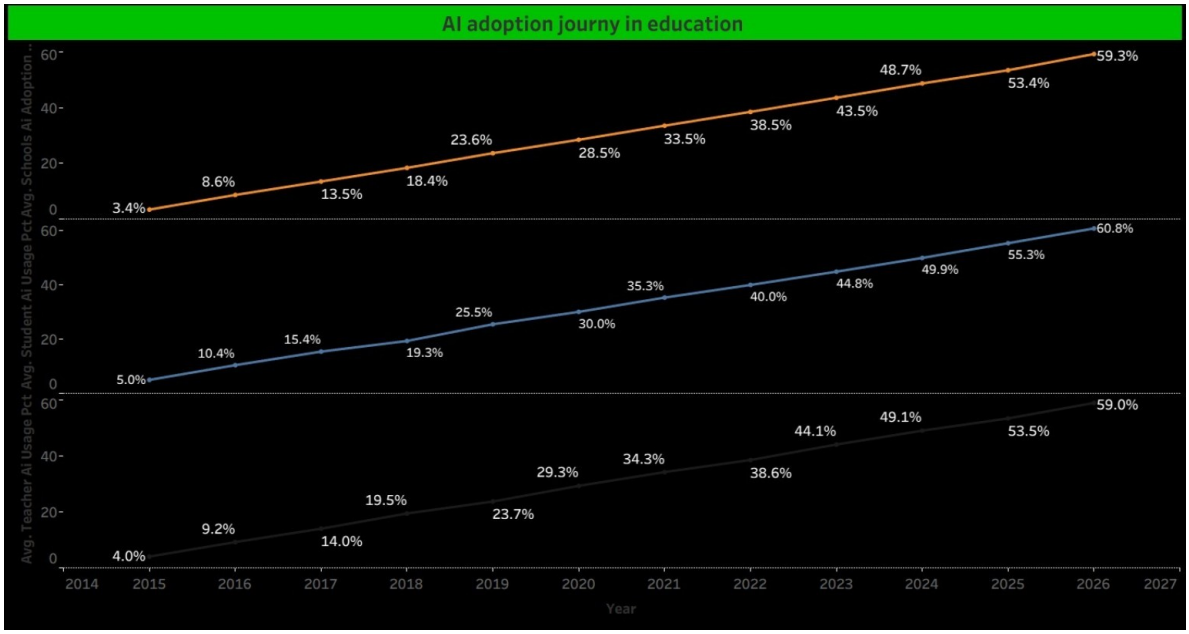


Figure: AI adoption journey in education

5. How does average daily AI usage vary by country?

Visualization: The country comparison shows average daily usage ranging from 30.2 minutes in the United States to 35.5 minutes in Canada among the countries shown.

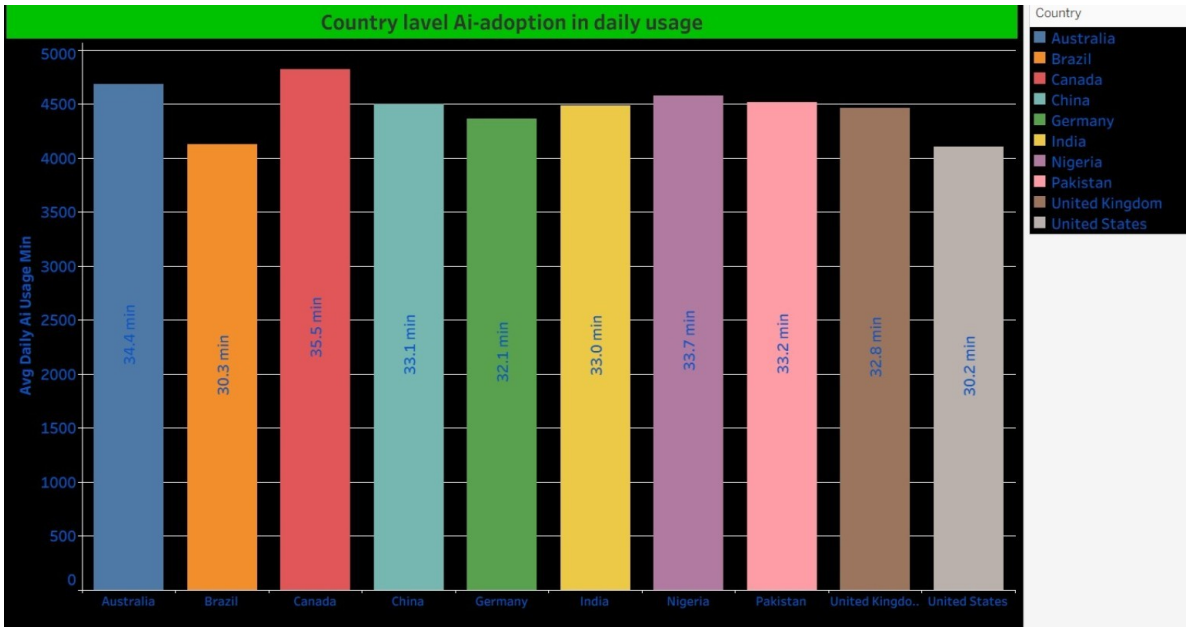


Figure: Country-level AI adoption in daily usage

Note: Five supplied visualizations have been included. Additional business questions and visualizations can be added if further charts are provided.

Note: Min 8 business question and visualisations Required inform of above

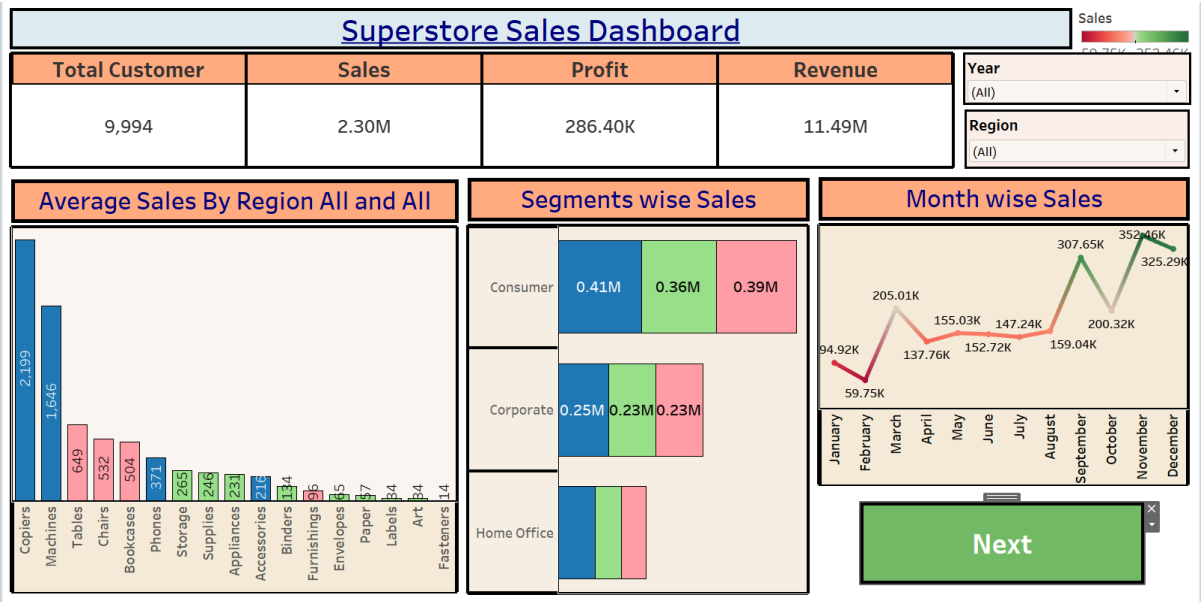
Dashboard Design

Creating an effective dashboard involves thoughtful design to ensure that the presented information is clear, relevant, and easily understandable for the intended audience. Here are some key principles and best practices for dashboard design

Activity 1: Interactive and visually appealing dashboards

Creating interactive and visually appealing dashboards involves a combination of thoughtful design, effective use of visual elements, and the incorporation of interactive features. Here are some tips to help you design dashboards that are both visually appealing and engaging for users so take care of below points

- Clear and Intuitive Layout
- Use Appropriate Visualizations
- Colour and Theming
- Interactive Filters and Slicers
- Drill-Down Capabilities
- Responsive Design
- Custom Visuals and Icons
- Use of Infographics



Note: Highlight the major outcomes in form of bullet points

Sample:

Project Dashboard: Global AI Adoption in Education

• Geographic and regional coverage: The dashboard presents country-level and regional comparisons, helping users explore how AI adoption and usage vary around the world.

• Daily usage by country: Average daily AI usage ranges from 30.2 minutes in the United States to 35.5 minutes in Canada across the ten countries displayed.

• Rural–urban comparison: Both rural and urban AI usage rise over time. The urban series remains above the rural series in the displayed years.

• Adoption journey: Student, school, and teacher AI usage all show sustained growth between 2015 and 2026, reaching approximately 59–61% by 2026.

• Adoption level breakdown: Low adoption is the largest category (680), followed by moderate adoption (473) and high adoption (207).

• Interactive exploration: Dashboard filters and navigation allow viewers to move between the country, regional, rural–urban, and adoption-trend views.

Dashboard screenshots:

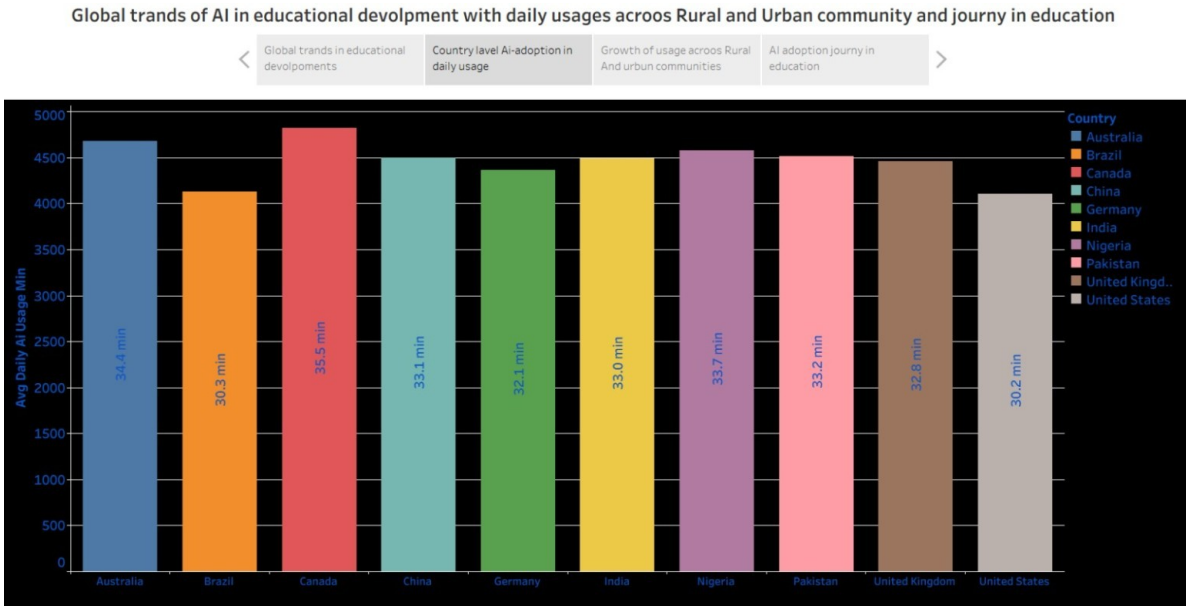


Figure 1. Global AI adoption dashboard overview (combined view).

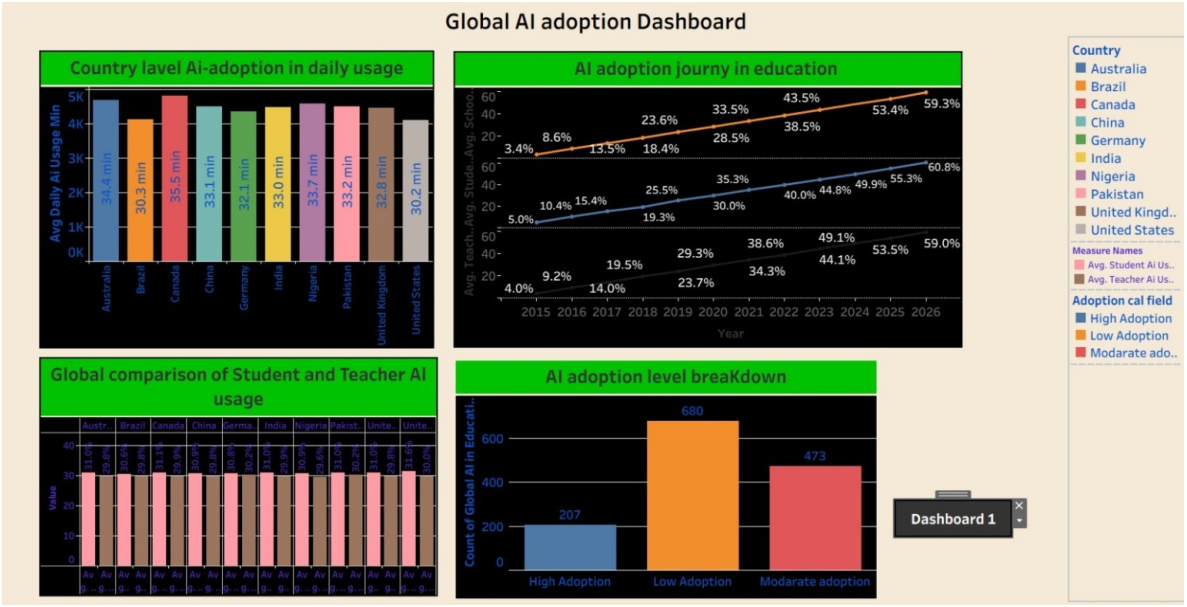


Figure 2. Global AI adoption dashboard – Dashboard 1.

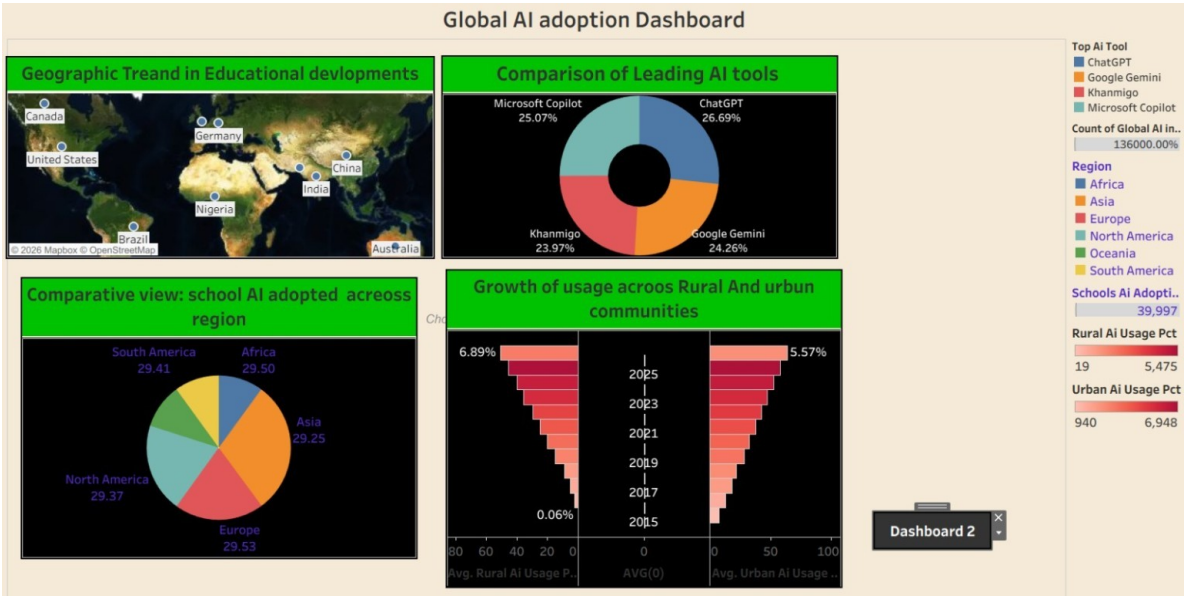


Figure 3. Global AI adoption dashboard – Dashboard 2.

Activity 2

Publish Dashboard on Tableau Public and Paste the Dashboard Public link below:

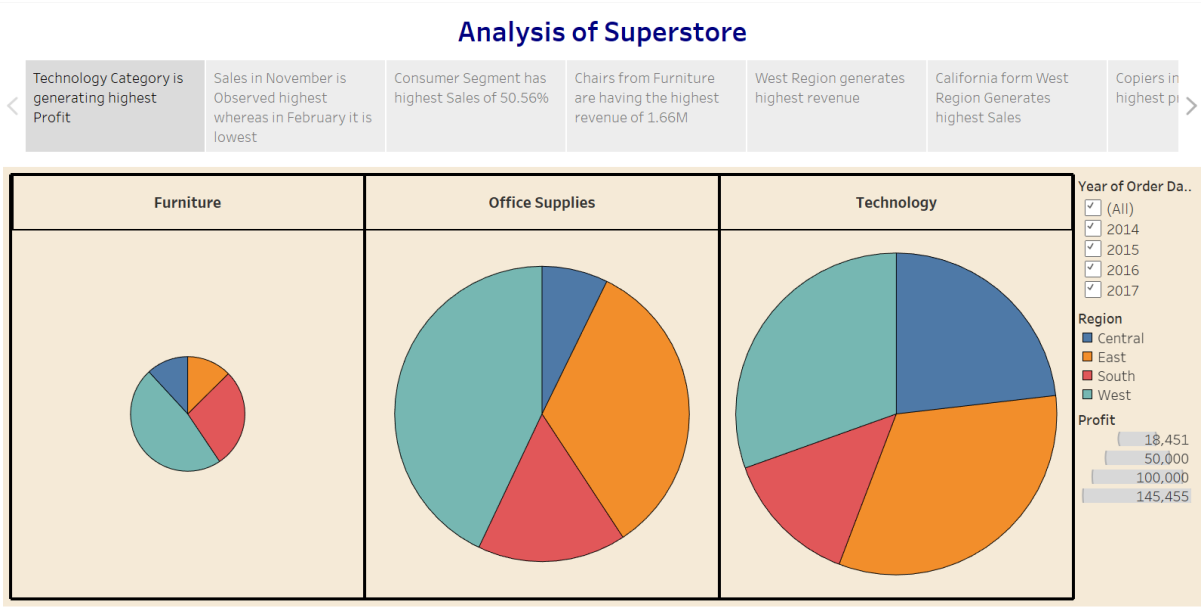
Dashboard publishing link:

[Insert your published Tableau Public Dashboard URL here]

Story

By using stories in Tableau, you can effectively communicate complex data in a way that is both interactive and engaging, making it easier for the audience to follow along and understand the insights. It's a tool for **data storytelling**, allowing you to present insights in a cohesive, engaging way that takes viewers through a logical progression of findings or analyses.

In Tableau, **Story** is a feature that allows you to create a sequence of dashboards, visualizations, and text to present data insights in a cohesive and narrative-driven way. It's like a slideshow within Tableau that guides the audience through a series of data points, helping them understand key insights, trends, or outcomes of your analysis.



Observations:

Story: Global Trends of AI Adoption in Education

1. Adoption levels are uneven: The supplied breakdown contains 680 low-adoption records, compared with 473 moderate-adoption and 207 high-adoption records. This points to substantial room for broader AI integration.

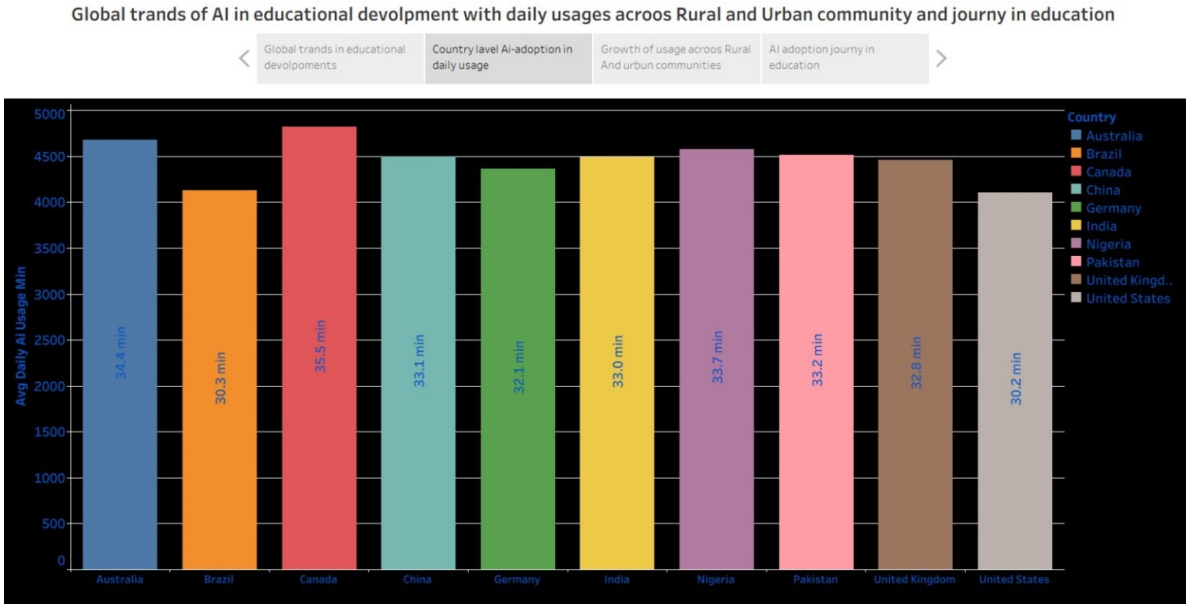
2. Daily usage differs by country, but remains within a relatively narrow range: The ten countries shown average approximately 30.2 to 35.5 minutes of daily AI usage. Canada is highest in this comparison, while the United States is lowest.

3. Rural and urban usage are both increasing: The time-series comparison shows growth in both communities, with urban usage consistently higher than rural usage across the displayed years.

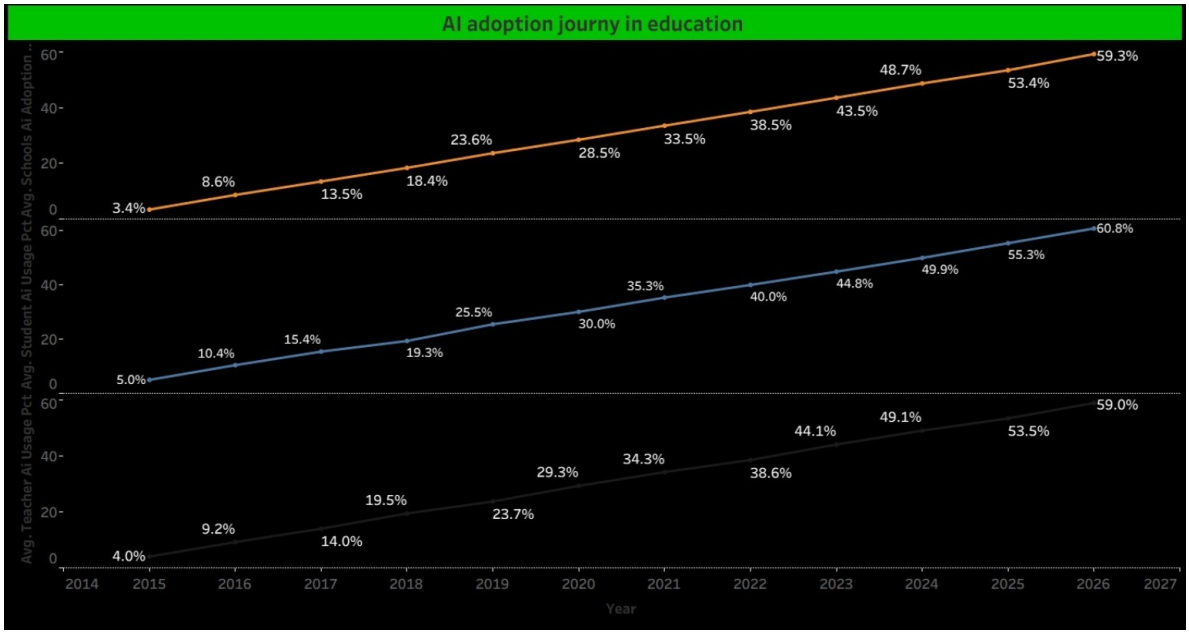
4. AI use in education is rising: The student, school, and teacher measures all trend upward from 2015 to 2026, reaching around 59–61% at the end of the period.

5. Regional comparison: The displayed regional values are close to one another (approximately 29.25–29.53), suggesting a relatively balanced distribution in this particular measure.

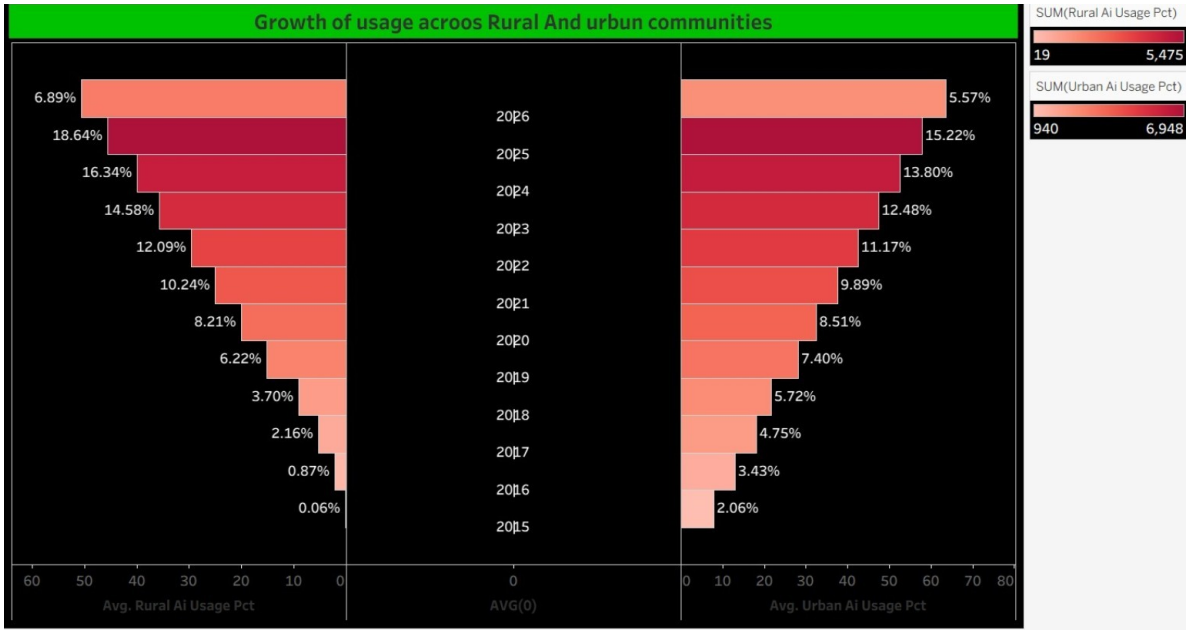
Story visuals:



Story point 1. Global dashboard overview.



Story point 2. AI adoption journey in education.



Story point 3. Growth of usage across rural and urban communities.

Activity 2:

Publish Story on Tableau Public and Paste Public link below

[Insert your published Tableau Public Story URL here]